

Fourth Presentation

Vinícius BAIÃO PIRES - Nahla EL HARRAK

2026

- 1 Architecture Overview
- 2 GNN Encoder
- 3 HMCN-F Head
- 4 Loss Function
- 5 Evaluation Metrics
- 6 Validation Track: Classical ML

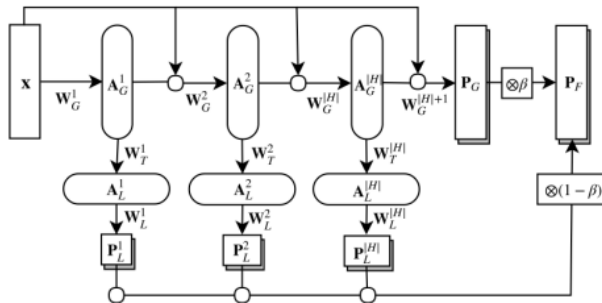


Figure: Wehrmann et al., ICML 2018

① Initial projection

$$A_G^0 = \text{ReLU}(\text{BN}(W_G^0 x + b_G^0)) \in \mathbb{R}^{128}$$

② Level 1 — fine labels (138)

$$A_G^1 = \text{ReLU}(\text{BN}(W_G^1 [A_G^0 \oplus x])) \quad (\text{input reuse})$$

$$P_{L1} = \sigma(W_{L1} \text{ReLU}(W_{T1} A_G^1)) \in [0, 1]^{138}$$

③ Level 2 — metacategories (12)

$$A_G^2 = \text{ReLU}(\text{BN}(W_G^2 [A_G^1 \oplus x]))$$

$$P_{L2} = \sigma(W_{L2} \text{ReLU}(W_{T2} A_G^2)) \in [0, 1]^{12}$$

④ Global output & final prediction

$$P_G = \sigma(W_G A_G^2) \in [0, 1]^{150}$$

$$P_F = \beta [P_{L1} \parallel P_{L2}] + (1 - \beta) P_G, \quad \beta = 0.5$$

Full grid (24 configs)

Parameter	Values
global_dim	{64, 128}
local_dim	{32, 64}
dropout	{0.50, 0.60, 0.70}
lr	$\{10^{-3}, 5 \times 10^{-4}\}$
weight_decay	$\{10^{-4}, 10^{-3}\}$
lambda_viol	{0.05, 0.1}

Final configuration

Parameter	Value
global_dim / local_dim	128 / 64
dropout	0.47
lr	10^{-4}
weight_decay	10^{-4}
λ_{viol}	0.1
β (local/global)	0.5
scheduler	CosineWarmRestarts

Before

```
META_CATEGORIES = {  
  "floral": [  
    "floral", "rose",  
    "jasmin", ...  
  ],  
  "fruity": [  
    "fruity", "apple",  
    "banana", ...  
  ],  
  ...  
}
```

After

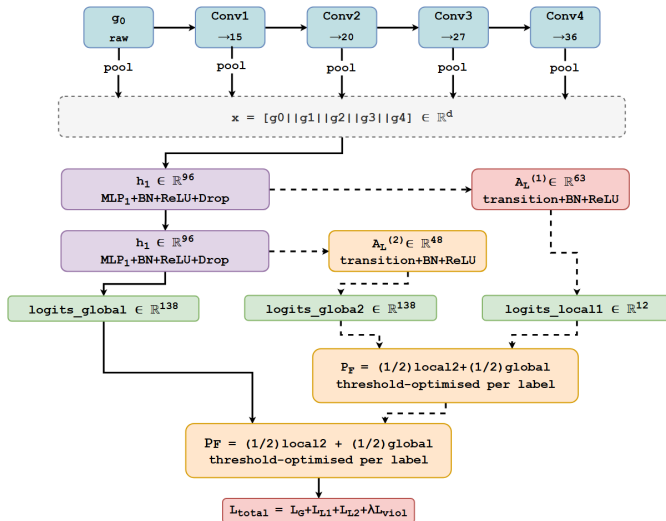
```
META_CATEGORIES = {  
  "macro_floral": [  
    "floral", "rose",  
    "jasmin", ...  
  ],  
  "macro_fruity": [  
    "fruity", "apple",  
    "banana", ...  
  ],  
  ...  
}
```

- Input: a **SMILES string** representing a small molecule
- **GNN encoder**: $4\times$ GCNConv layers with SELU activation; each layer's pooled output is *retained* (skip connections)
- **HMCN-F head**: two-level hierarchy — 12 macro-categories $\subset$ 138 fine labels
- Three output heads: logits_global, logits_local1, logits_local2
- Final prediction: β -weighted blend of global and local-2 heads ($\beta = 0.5$)

Dataset

Source	Good Scents + Leffingwell
Molecules	≈ 4983
Labels (fine)	138
Macro-categories	12
Split	80/10/10 (stratified)

Architecture Diagram



Four GCNConv layers with hidden dims $15 \rightarrow 20 \rightarrow 27 \rightarrow 36$:

$$h_i^{(k)} = \phi \left(W^{(k)} \tilde{D}^{-\frac{1}{2}} \tilde{A} \tilde{D}^{-\frac{1}{2}} H^{(k-1)} \right)_i$$

where ϕ is SELU. **Global add pooling** after every layer:

$$g_k = \sum_{i=1}^n h_i^{(k)}, \quad k \in \{0, 1, 2, 3, 4\}$$

The five pooled vectors are **concatenated**:

$$x = [g_0 \| g_1 \| g_2 \| g_3 \| g_4] \in \mathbb{R}^d$$

Global Flow — Two-Level Representation

The HMCN-F head maps $x \in \mathbb{R}^d$ through two successive FC blocks.

Level-1 activation h_1

$$h_1 = \text{ReLU}(\text{BN}_1(W_{G,1}x + b_{G,1})) \xrightarrow{\text{Drop}(0.47)} h_1 \in \mathbb{R}^{96}$$

(global_mlp1 $\rightarrow$ global_bn1 $\rightarrow$ ReLU $\rightarrow$ global_drop1)

Level-2 activation h_2

$$h_2 = \text{ReLU}(\text{BN}_2(W_{G,2}h_1 + b_{G,2})) \xrightarrow{\text{Drop}(0.47)} h_2 \in \mathbb{R}^{63}$$

(global_mlp2 $\rightarrow$ global_bn2 $\rightarrow$ ReLU $\rightarrow$ global_drop2)

Global output head

$$\text{logits_global} = W_{G,\text{out}} h_2 + b_{G,\text{out}} \in \mathbb{R}^{138}$$

Predicts all 138 fine-grained odour labels simultaneously.

Each local flow branches off the global flow via a *transition layer*.

Local Level-1 (branches from h_1)

$$A_L^{(1)} = \text{ReLU}\left(\text{BN}_T^{(1)}\left(W_T^{(1)}h_1\right)\right) \in \mathbb{R}^{48}$$

$$\text{logits_local1} = W_L^{(1)}A_L^{(1)} \in \mathbb{R}^{12}$$

Predicts 12 macro-category groups.

h_1 is a 96-dim, earlier representation — coarser, ideal for broad odour families.

Local Level-2 (branches from h_2)

$$A_L^{(2)} = \text{ReLU}\left(\text{BN}_T^{(2)}\left(W_T^{(2)}h_2\right)\right) \in \mathbb{R}^{63}$$

$$\text{logits_local2} = W_L^{(2)}A_L^{(2)} \in \mathbb{R}^{138}$$

Predicts all 138 fine-grained labels.

h_2 is 63-dim, more compressed — captures cross-category dependencies.

At inference, the three outputs are fused:

$$P_F = \beta \cdot \text{logits_local2} + (1 - \beta) \cdot \text{logits_global}, \quad \beta = 0.5$$

Role of each head

Head	Shape	Used for
logits_global	$\mathbb{R}^{138}$	global loss + P_F
logits_local1	$\mathbb{R}^{12}$	macro BCE loss only
logits_local2	$\mathbb{R}^{138}$	local-2 loss + P_F

Per-label threshold optimisation

$$\tau_j^* = \arg \max_{\tau} \text{F1}(y_j, 1[\hat{p}_j \geq \tau])$$

Loss Function — Four Components

$$\mathcal{L}_{\text{total}} = \mathcal{L}_G + \mathcal{L}_{L_1} + \mathcal{L}_{L_2} + \lambda \mathcal{L}_{\text{viol}}$$

$\mathcal{L}_G$ — Global BCE (pos-weighted)

$$\mathcal{L}_G = \mathcal{E}_{\text{pos}}(\text{logits_global}, y_{138})$$

$\mathcal{L}_{L_2}$ — Local-2 BCE (pos-weighted)

$$\mathcal{L}_{L_2} = \mathcal{E}_{\text{pos}}(\text{logits_local2}, y_{138})$$

$\mathcal{L}_{L_1}$ — Macro BCE (unweighted)

$$y_p^{(\text{macro})} = \max_{c \in \text{children}(p)} y_c$$

$$\mathcal{L}_{L_1} = \mathcal{E}(\text{logits_local1}, y_{12}^{(\text{macro})})$$

$\mathcal{L}_{\text{viol}}$ — Hierarchy violation penalty

For validated pairs $\mathcal{V}$ with $P(\text{parent}|\text{child}) \geq 0.6$:

$$\mathcal{L}_{\text{viol}} = \frac{1}{|\mathcal{V}|} \sum_{(i_n, i_p) \in \mathcal{V}} \max(0, \hat{p}_{i_n} - \hat{p}_{i_p})^2$$

The models are evaluated at **two levels**:

- 1 **Fine level** (138 labels)
- 2 **Macro level** (12 groups)

Thresholds are found on **validation set**, applied to **test set only**.

Metric categories

Category	Metrics
Ranking	ROC-AUC, PR-AUC
Threshold	F1 (macro/micro/samples)
Per-class	Precision, Recall
Hierarchical	Violation Rate
12-group	Balanced Acc., Sensitivity, Specificity, Co-occurrence

ROC-AUC (Area Under the ROC Curve)

For label j with scores $\hat{p}_j$:

$$\text{TPR}(\tau) = \frac{TP(\tau)}{TP(\tau) + FN(\tau)}, \quad \text{FPR}(\tau) = \frac{FP(\tau)}{FP(\tau) + TN(\tau)}$$

PR-AUC (Area Under the Precision-Recall Curve)

$$\text{Prec}(\tau) = \frac{TP(\tau)}{TP(\tau) + FP(\tau)}, \quad \text{Rec}(\tau) = \frac{TP(\tau)}{TP(\tau) + FN(\tau)}$$

All require binarised predictions $\hat{y}_j = 1[\hat{p}_j \geq \tau_j^*]$.

Precision, Recall, F1

$$P = \frac{TP}{TP + FP}, \quad R = \frac{TP}{TP + FN}$$
$$F_1 = \frac{2PR}{P + R} = \frac{2TP}{2TP + FP + FN}$$

Why threshold matters

At τ_j^* maximising F1:

$$FP = FN \implies P = R = F_1$$

12-Group Specific Metrics

Balanced Accuracy

For group p (binary classifier):

$$\text{Sens}_p = \frac{TP_p}{TP_p + FN_p}, \quad \text{Spec}_p = \frac{TN_p}{TN_p + FP_p}$$

$$\text{BalAcc}_p = \frac{\text{Sens}_p + \text{Spec}_p}{2}$$

Macro-average over groups with both classes present.

Hierarchical Violation Rate

$$\text{HVR} = \frac{\sum_{(i_n, i_p) \in \mathcal{V}} 1[\hat{y}_{i_n} = 1, \hat{y}_{i_p} = 0]}{\sum_{(i_n, i_p) \in \mathcal{V}} 1[\hat{y}_{i_n} = 1] + \epsilon}$$

Measures post-threshold logical consistency.

Label Co-occurrence Consistency

Compares predicted co-occurrence $\hat{C}$ vs. ground-truth C for 12 macro-groups:

$$\text{LCC} = \text{mean} |C / (C_{\max} + \epsilon) - \hat{C} / (\hat{C}_{\max} + \epsilon)|.$$

HMCN-F Results : Metacategory Level ($\lambda = 0.1$)

Per-label metrics (12 metacategories)

Label	ROC	PR	BalAcc	F1	Sens
floral	.846	.669	.770	.655	.687
fruity	.821	.793	.729	.728	.823
sweet	.743	.644	.663	.613	.788
woody	.825	.609	.707	.525	.553
green	.775	.723	.674	.689	.870
spicy	.758	.495	.664	.459	.606
animal_musk	.778	.434	.708	.489	.541
earthy	.770	.528	.692	.505	.522
citrus	.845	.455	.695	.451	.451
chemical	.780	.726	.697	.677	.789
gourmand	.828	.682	.750	.601	.731
powdery_amb	.802	.446	.656	.454	.422
Macro	.798	.600	.700	.571	.649

Instance-based

Metric	Value
Exact Match	0.059
Instance F1	0.592

HMCN-specific

Metric	Value
Hier. Violation Rate	0.201
Co-occ. Consistency	0.136

HVR: 19.1% of child positive predictions violate the hierarchy.

Co-occurrence: worst pair floral and sweet (error 0.41).

HMCN-F Results : Fine Label Level (138 labels, $\lambda = 0.1$)

Top 20 fine labels by PR AUC

Label	ROC	PR	F1
fruity	.815	.753	.677
sulfurous	.974	.739	.718
odorless	.858	.633	.603
fatty	.871	.591	.571
green	.740	.567	.530
floral	.818	.562	.533
onion	.957	.554	.549
woody	.845	.526	.483
sweet	.721	.496	.502
waxy	.890	.463	.565
meaty	.927	.452	.554
alcoholic	.887	.447	.326
garlic	.932	.429	.419
apple	.879	.407	.389
roasted	.913	.398	.476
coffee	.865	.384	.372
vanilla	.862	.384	.431
ethereal	.864	.359	.387
citrus	.842	.356	.417

Macro averages (all 138 labels)

Metric	Value
ROC AUC	0.754
PR AUC	0.155
Bal. Acc	0.586
F1 (τ^*)	0.152
Sensitivity	0.224
Specificity	0.948

Binary Relevance

Train one independent binary classifier per label:

$$\hat{y}_k = f_k(x), \quad k = 1, \dots, 12$$

Labels are treated as independent.

5 classifiers tested

	Classifier	Grid
LR	Logistic Regression	$C \in \{0.01, \dots, 100\}$
RF	Random Forest	$n_est \in \{100, 300\}$ $max_feat \in \{\text{sqrt}, \text{log2}\}$
XGB	XGBoost	$n_est \in \{100, 300\}$ $max_depth \in \{3, 6\}$
SVM	SVM (RBF)	$C \in \{0.1, 1, 10\}$
KNN	k -NN	$k \in \{5, 11, 21\}$ $metric \in \{\text{euclid}, \text{cosine}\}$

BR Results : Macro-Averaged Metrics (12 Metacategories)

Model	Bal.Acc	F1	ROC	PR	Sens	Spec	ExM	InstF1
LR	0.693	0.624	0.802	0.683	0.709	0.677	0.037	0.654
RF	0.694	0.620	0.804	0.697	0.644	0.744	0.031	0.655
XGB	0.696	0.623	0.792	0.683	0.692	0.700	0.022	0.641
SVM	0.707	0.636	0.803	0.684	0.690	0.725	0.034	0.663
KNN	0.696	0.624	0.775	0.620	0.681	0.711	0.034	0.654

BR vs HMCN-F : Comparison (12 Metacategories)

Model	ROC	PR	Bal.Acc	F1	Sens	Spec	ExM	InstF1
BR+LR	0.802	0.683	0.693	0.624	0.709	0.677	0.037	0.654
BR+RF	0.804	0.697	0.694	0.620	0.644	0.744	0.031	0.655
BR+XGB	0.792	0.683	0.696	0.623	0.692	0.700	0.022	0.641
BR+SVM	0.803	0.684	0.707	0.636	0.690	0.725	0.034	0.663
BR+KNN	0.775	0.620	0.696	0.624	0.681	0.711	0.034	0.654
HMCN-F	0.798	0.600	0.700	0.571	0.649	0.752	0.059	0.592

Level	Metric	Value
<i>Fine (138 labels)</i>	ROC AUC	0.8702
	PR AUC	0.2726
	F1 (macro)	0.2584
	Hier. Violation Rate	0.2109
<i>Meta (12 groups)</i>	ROC AUC	0.8021
	PR AUC	0.5862
	F1 (macro)	0.5742
	Balanced Accuracy	0.7173
	Sensitivity (macro)	0.5929
	Specificity (macro)	0.8416
	Hier. Violation Rate	0.2109
	Label Co-occ. Consistency	0.0599
—	Loss	2.0741

Level	Metric	Value
<i>Fine (138 labels)</i>	ROC AUC	0.8691
	PR AUC	0.2729
	F1 (macro)	0.2609
	Hier. Violation Rate	0.1863
<i>Meta (12 groups)</i>	ROC AUC	0.8045
	PR AUC	0.5900
	F1 (macro)	0.5714
	Instance-F1	0.5382
	Balanced Accuracy	0.7133
	Sensitivity (macro)	0.5886
	Specificity (macro)	0.8380
	Hier. Violation Rate	0.1863
	Label Co-occ. Consistency	0.0641
—	Loss	2.0864

Level	Metric	Value
<i>Fine (138 labels)</i>	ROC AUC	0.8695
	PR AUC	0.2714
	F1 (macro)	0.2560
	Hier. Violation Rate	0.2106
<i>Meta (12 groups)</i>	ROC AUC	0.8045
	PR AUC	0.5893
	F1 (macro)	0.5693
	Instance-F1	0.5344
	Balanced Accuracy	0.7106
	Sensitivity (macro)	0.5685
	Specificity (macro)	0.8527
	Hier. Violation Rate	0.2106
	Label Co-occ. Consistency	0.0607
—	Loss	2.0711

Level	Metric	Value
<i>Fine (138 labels)</i>	ROC AUC	0.8660
	PR AUC	0.2836
	F1 (macro)	0.2670
	Hier. Violation Rate	0.1914
<i>Meta (12 groups)</i>	ROC AUC	0.8046
	PR AUC	0.6073
	F1 (macro)	0.5790
	Instance-F1	0.5509
	Balanced Accuracy	0.7169
	Sensitivity (macro)	0.5916
	Specificity (macro)	0.8422
	Hier. Violation Rate	0.1914
	Label Co-occ. Consistency	0.0536
—	Loss	2.2247